

WE DEFINE
FUTURE IMPACT

TECHNOLOGY
SOFTWARE
CONSULTING

P3 BUSINESS
AS UNUSUAL



TMS: Database Workshop

Preserving Engineering Knowledge through Visualization & Collaboration.

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Nagel Group

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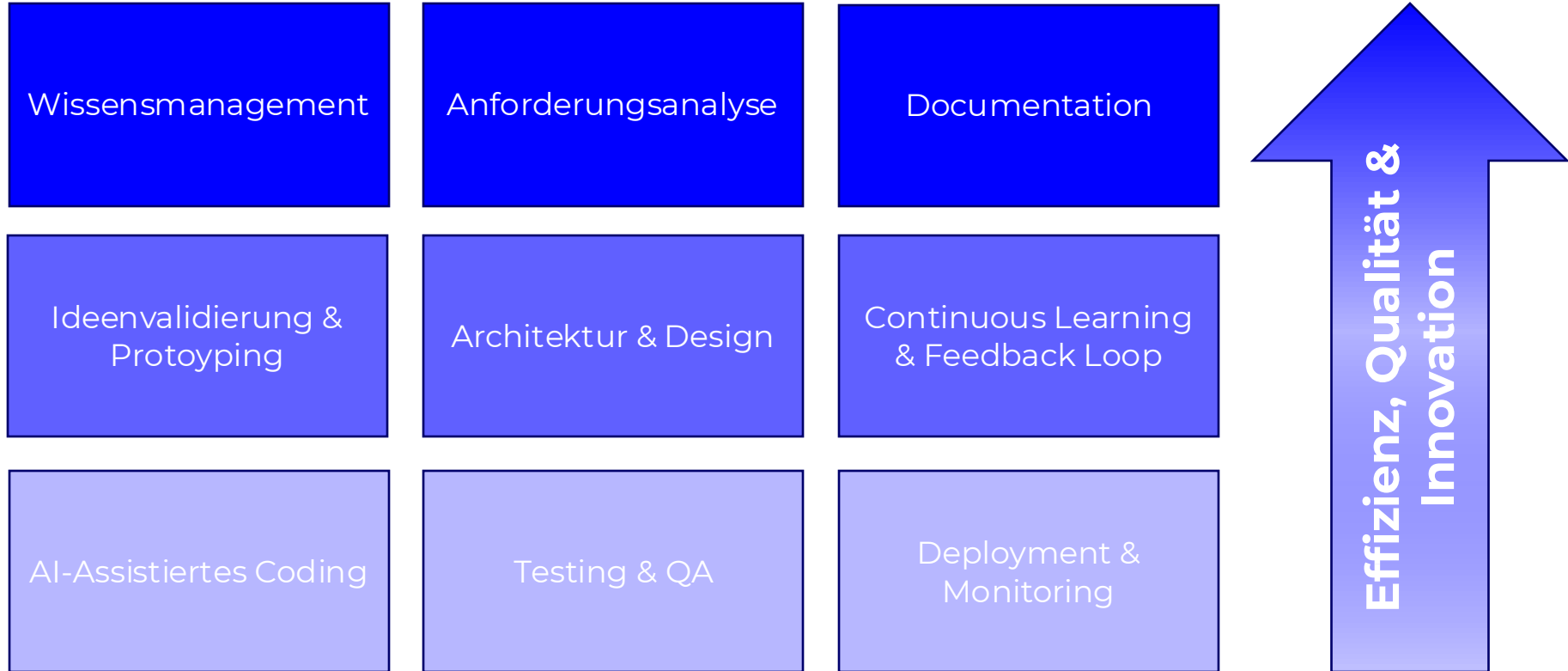
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Germany



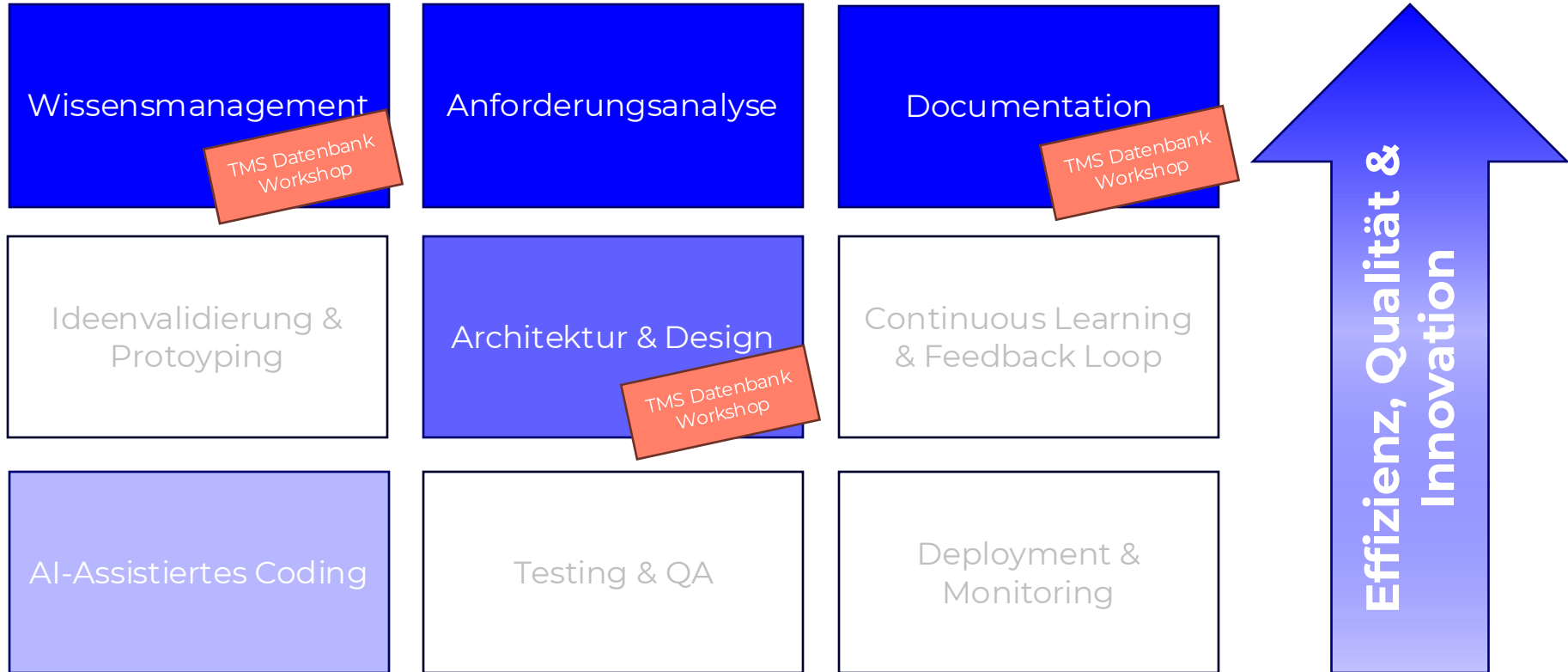
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AI-driven Software Development Process (Blueprint)



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Workshop-Ziele

Für Nagel

Kopfwissen
persistieren

Engpässe beseitigen

Begriffe
demystifizieren

Innovation
ermöglichen

Für P3

Architektur verstehen

Effizientere
Refinements

Handlungsfähig
werden

Beraten, nicht
verlängerte
Werkbank

Für Entwickler

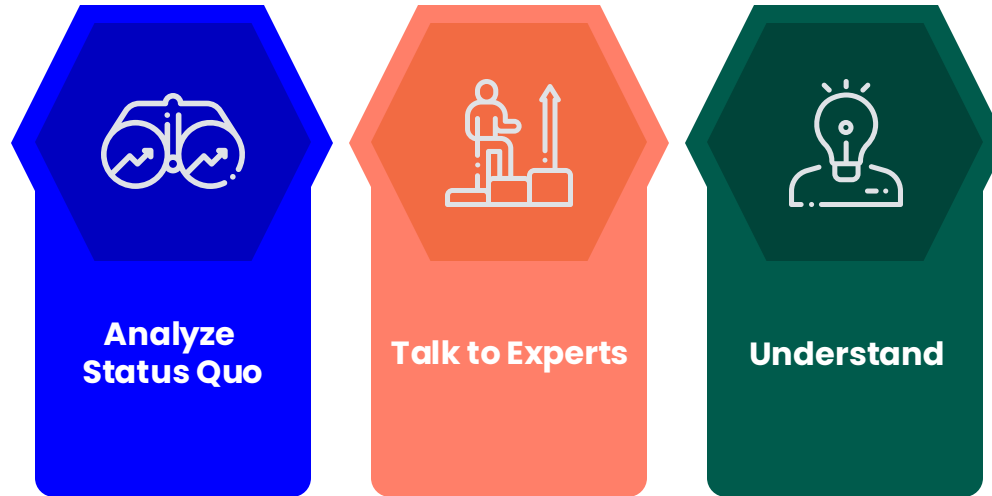
Entlastung

Fokus auf wertvolle
Arbeit

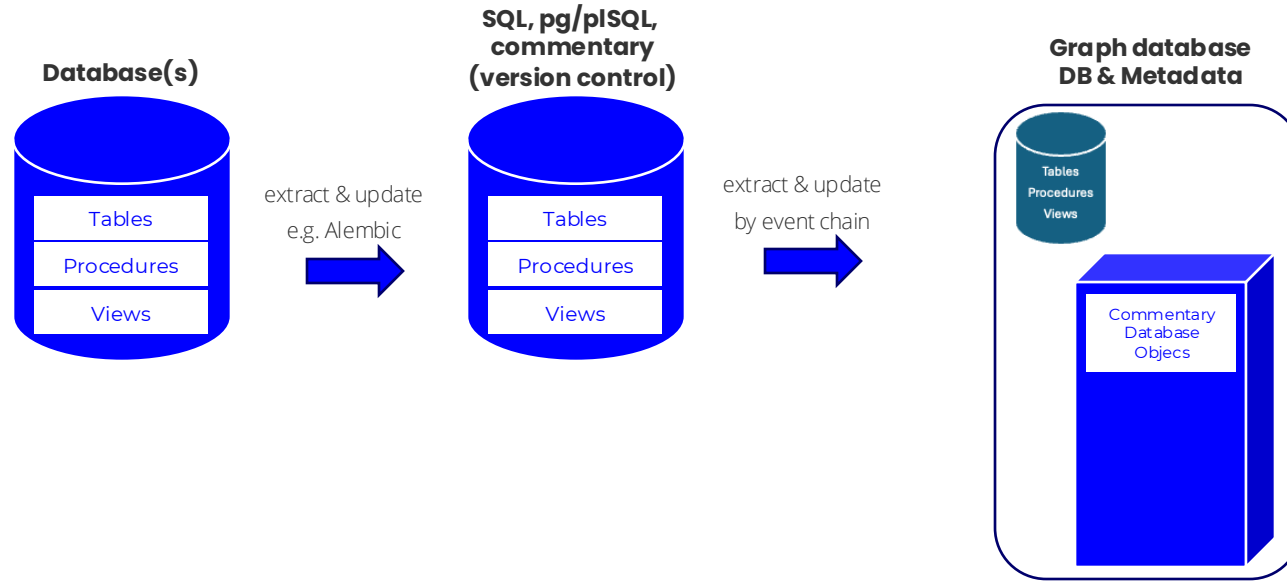
Höhere Velocity,
höhere Qualität

Vorstufe des AI-Driven SW Prozess

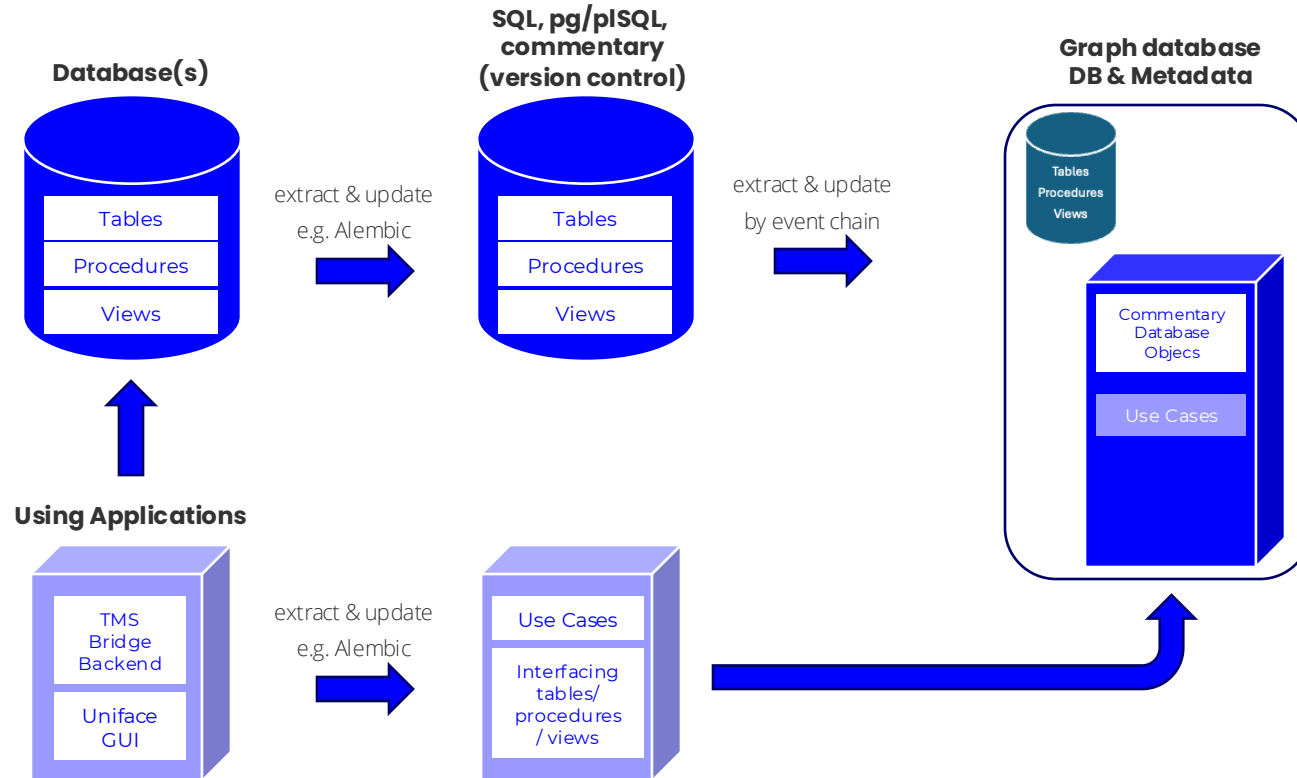
3 STEPS to fully understand and document for future development and operation



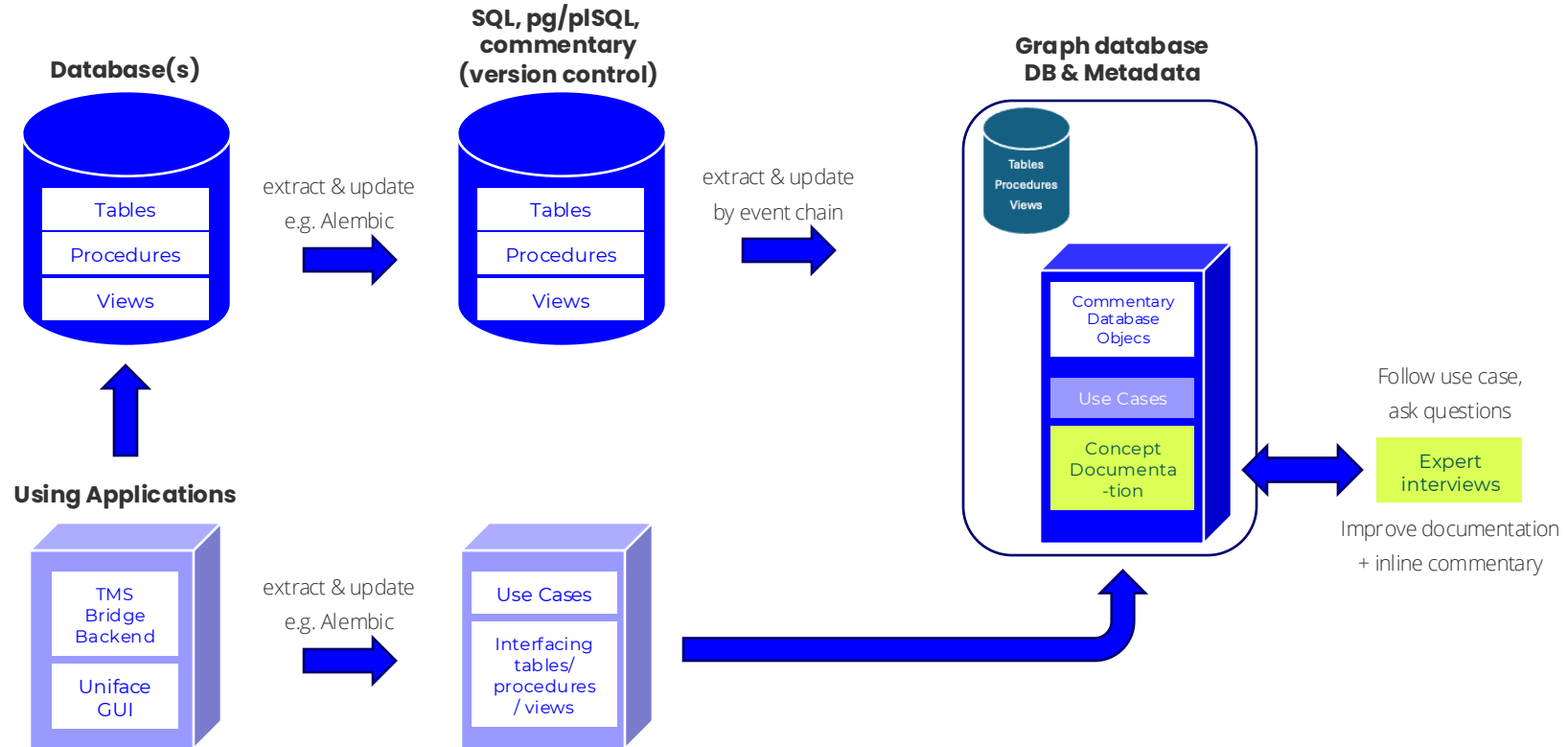
Iterative approach



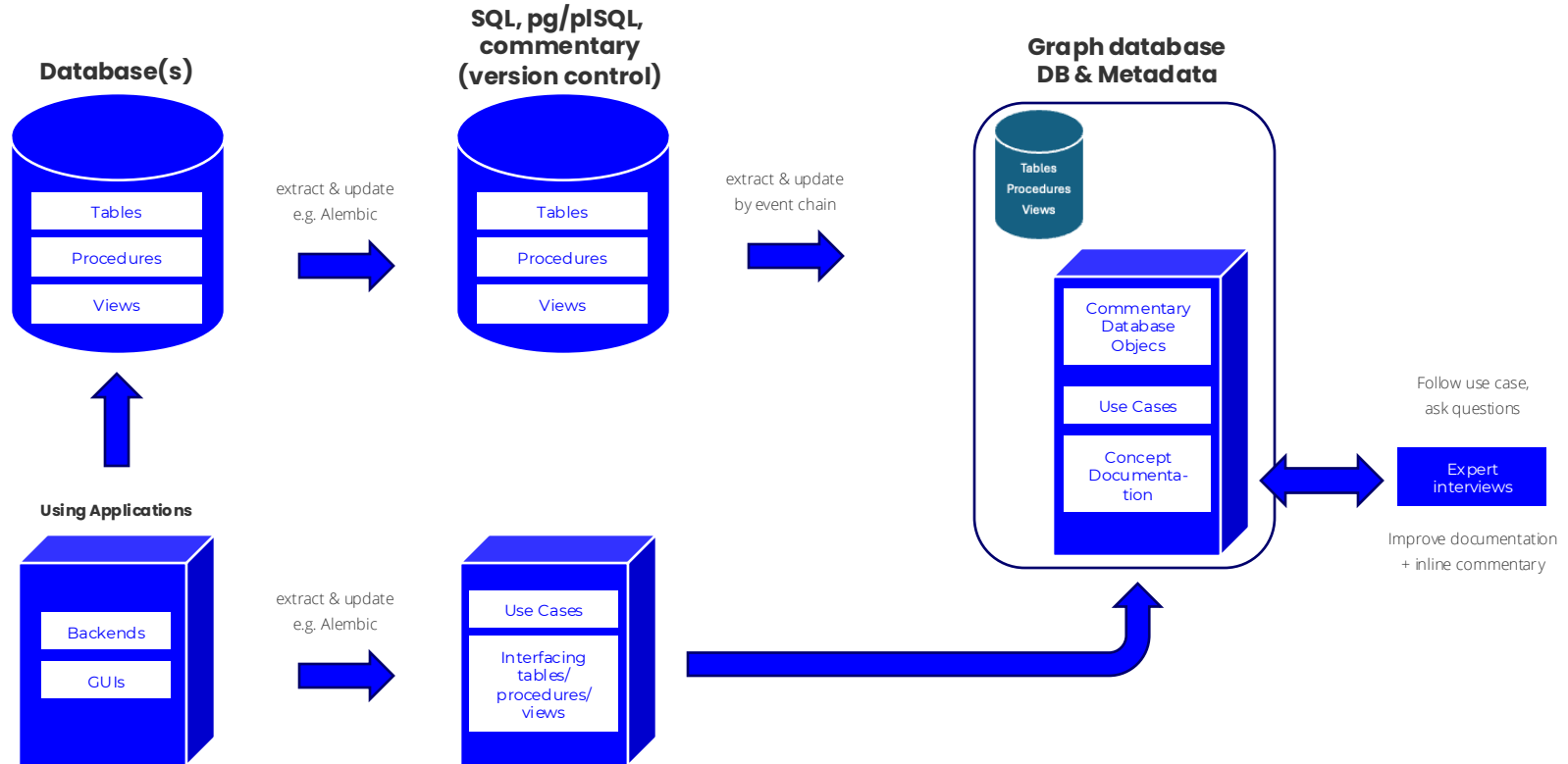
Iterative approach



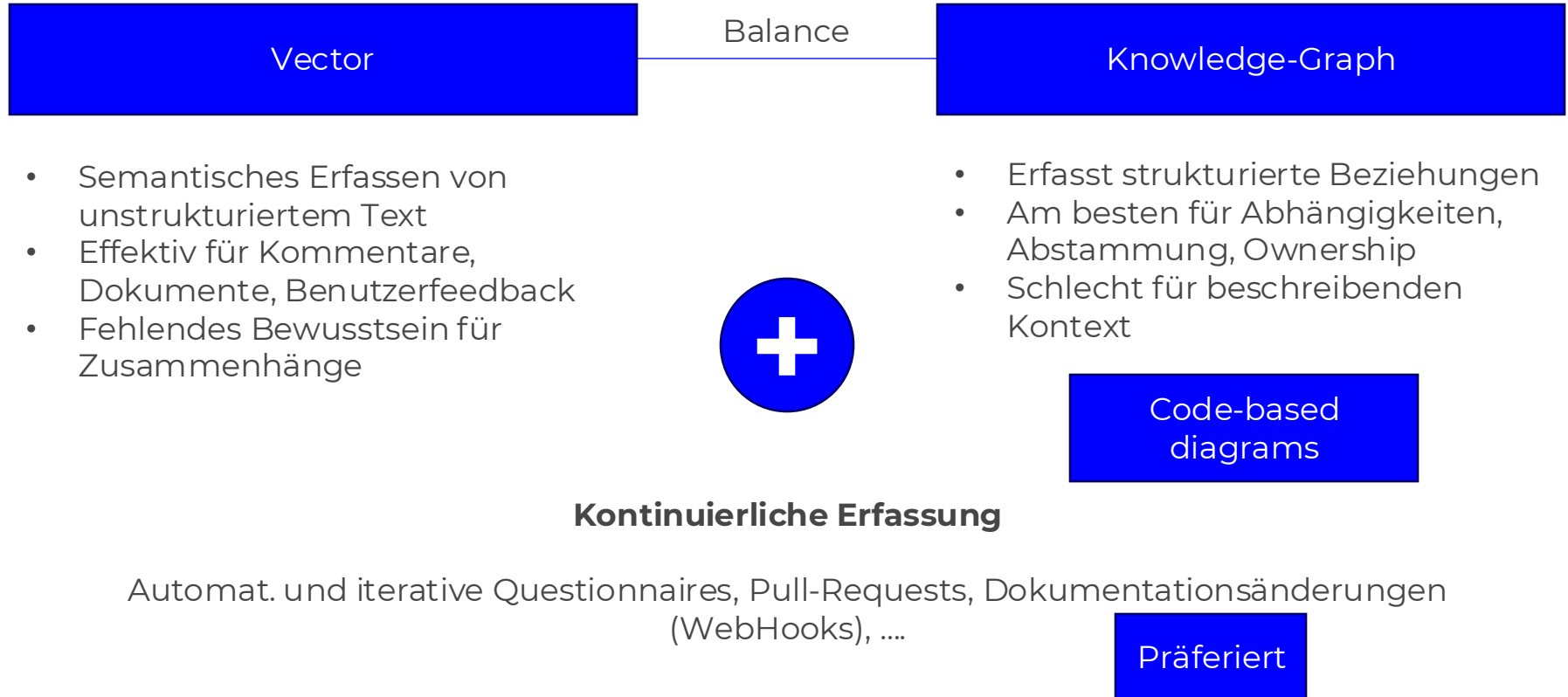
Iterative approach



Iterative approach



Wie persistieren wir Wissen?



AI-driven Software Development Process (Use Cases)

Dokumentation für alle zugänglich => Problem:
dynamischer Inhalt, schnell veraltet =>
Knowledge wäre flexibel und die statische Doku
nur ein Output des „Kerns“

Code-generierung z.B. für TMS-Bridge =>
Nutzen des Wissens wie Glossar, APIs und
bestehender Code, um Datenbank Features
anzubinden + HITL als QA-Gate

Refinements beschleunigen, effizienter, alle
benötigen Quellen immer zur Hand, Output:
Features/Stories in „unserer Sprache“, hohe
Qualität, INVEST

Prototyping auf Basis des Wissens (APIs,
Datenbank, Prozesse, UI Design System) =>
funktionale Prototypen statt Figma-UI-only
= Need gleich validiert

WORKSHOP AGENDA

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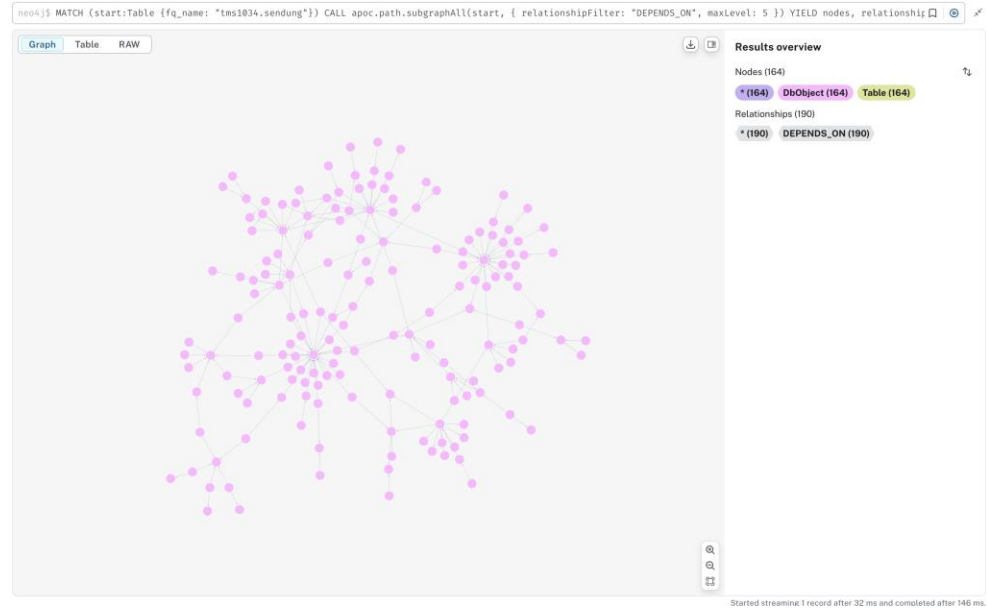
Workshop Day 1: Orientation & Discovery

Morning

- Introduction & Why It Matters
- Graph walkthrough: schema, functions, dependencies
- Missing context: where human input is key

Afternoon

- Domain Deep Dive (e.g., Disposition, Abfertigung, Mobile, Messpunkte)



Fokus: TA, Shipment, Driver-Terminal-Schnittstelle, Tourpunkte, Tourenoptimierung => bekannte, bereits genutzte Domains

Workshop Day 2: Logic Threads & Concepts

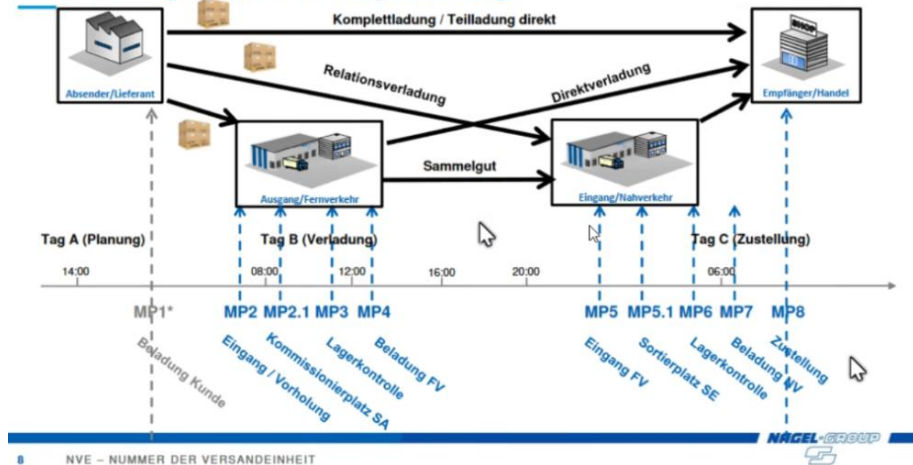
Morning

- Scenario-based tracing (e.g., TA-Planungsprozess, Mobile-Scanning)
- Engineer pairs explore and annotate flows

Afternoon

- Introduce Concept Nodes
- Map business concepts to logic nodes
- Add tags and context

NVE-Messpunkte der Nagel-Group



Workshop Day 3: Ownership & Review

Morning

- Assign ownership of graph sections
- Plan breakout follow-ups for gaps
- LOP / Next Steps
- Regeltermine + iterativer Modus

Afternoon

- Final team review and feedback

Christian

- Performanceprobleme in PG vs. Oracle => Graph als Basis für Analyse => Beispiel: View ist 200x langsamer in PG
- = akute Themen

Engineer Prompt Cards

Function Nodes

- What does this do in business terms?
- Any legacy behavior or risks?
- Is this still relevant in New Dispo?

Concept Mapping

- What process does this logic support?
- Who relies on this output?

Reporting

- Haben alle techn. Komponenten ein Business-Knowledge Mapping?
- Kennzahlen ableiten (X von Y techn. Komponenten stehen im Business-Kontext)
- Komplexitäts-/Completeness Metriken je Domäne
- Abhängigkeiten zwischen Domänen nicht nur techn. Komponenten

Workshop Goals

AB HIER INHALT NOCH ZU PRÜFEN

Workshop Goals

Goals

- Capture undocumented knowledge from engineers
- Enrich technical database objects with business and functional context
- Assign ownership
- Establish an update process

Bonus

- Enable LLM-powered Q&A and documentation

WORKSHOP VORBEREITUNGEN

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Persist Knowledge and make it accessible

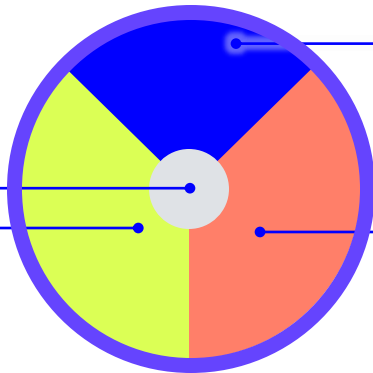
Knowledge Engine

The knowledge engine captures the database's schema and logic.

Domain Knowledge

Domain knowledge is linked with technical components in order to arrive at a holistic overall view of the solution.

Knowledge Graph



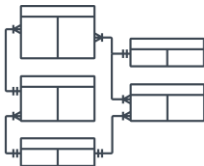
Code

Codebases are extracted automatically and their internal dependencies are recorded, visualized and made accessible.

Business Processes

Technical knowledge and domain knowledge come together in the process documentation to show transparently how value creation takes place.

Extracting Code Knowledge



Tables, Views, Relationships

Extracted from PostGres system catalogue.

Functions, Procedures

```
199 CREATE OR REPLACE FUNCTION test.detectstops() RETURNS void LANGUAGE 'plpgsql'
200 AS $$
201 BEGIN
202     PERFORM test_stop_at_beginning();
203     PERFORM test_stop_at_end();
204     PERFORM test_move_stop_beginning();
205     PERFORM test_two_stops();
206     PERFORM test_two_stops_with_gap();
207     PERFORM test_no_stops();
208     PERFORM test_two_short();
209     PERFORM test_long_stop_segment();
210 END; $$
```

Extracted from PostGres system catalogue and parsed with the AI Code Interpreter or manually.

Complete Relationship Mapping

All code components and their dependencies are fully documented and queryable.

Automation level: 100% (with AI)

Pre-Workshop Checklist

- Load schema into Knowledge Engine (tables, functions, views, procedures)
- Parse and map dependencies
- Liste aller Prozesse
- Group logic by domain/module
- Set up UX/UI Tool for browsing/editing during the workshop

Sync Architecture Council – 16.04.2023

Problem:

- Komplexes DB
- Blocker
 - Joachim Bottleneck
 - Dispo Projekt braucht TMS Zuarbeit
 - Selbst die vorübergehende Lösung Pretzel und TMS-Bridge funktioniert nicht
- Nürnberg Workshop: Initialer Aufsatz, danach continuous syncs
- Automatisierung, um schneller zu werden → KI

Agenda:

- Den Zyklus von Nils durchgehen
- Mappen der DB Functions in genutzt und nicht genutzt
- Ziel: Überführung in einfachere Prozesse
- Store procedures
- LLM Freigabe
- Code Base KPIs:
- Wichtig: Kommentare für die Aufarbeitung mit der KI

Nächste Schritte:

- Reinladen des Systemkatalogs: Matthias
- Repo zip: Matthias
- Nils Präsentation: Nils
- Aktan bereitet Präsi auf
- Komplettskan der DB: Matthias und Nils

Entscheidung Christian:

- KI → Go/ no Go?